

Avacnes2.5

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```
datos <-read.csv("test.csv")
```

```
##      Id SalePrice economico medio caro
## 1 1461  169277.1          1      0    0
## 2 1462  187758.4          0      1    0
## 3 1463  183583.7          1      0    0
## 4 1464  179317.5          1      0    0
## 5 1465  150730.1          1      0    0
## 6 1466  177151.0          1      0    0
```

##	Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LotShape	LandContour
## 1	1	60	RL	65	8450	Pave	<NA>	Reg	Lvl
## 2	2	20	RL	80	9600	Pave	<NA>	Reg	Lvl
## 3	3	60	RL	68	11250	Pave	<NA>	IR1	Lvl
## 4	4	70	RL	60	9550	Pave	<NA>	IR1	Lvl
## 5	5	60	RL	84	14260	Pave	<NA>	IR1	Lvl
## 6	6	50	RL	85	14115	Pave	<NA>	IR1	Lvl
##	Utilities	LotConfig	LandSlope	Neighborhood	Condition1	Condition2	BldgType		
## 1	AllPub	Inside	Gtl	CollgCr	Norm	Norm	1Fam		
## 2	AllPub	FR2	Gtl	Veenker	Feedr	Norm	1Fam		
## 3	AllPub	Inside	Gtl	CollgCr	Norm	Norm	1Fam		
## 4	AllPub	Corner	Gtl	Crawfor	Norm	Norm	1Fam		
## 5	AllPub	FR2	Gtl	NoRidge	Norm	Norm	1Fam		
## 6	AllPub	Inside	Gtl	Mitchel	Norm	Norm	1Fam		
##	HouseStyle	OverallQual	OverallCond	YearBuilt	YearRemodAdd	RoofStyle	RoofMatl		
## 1	2Story	7	5	2003	2003	Gable	CompShg		
## 2	1Story	6	8	1976	1976	Gable	CompShg		
## 3	2Story	7	5	2001	2002	Gable	CompShg		
## 4	2Story	7	5	1915	1970	Gable	CompShg		
## 5	2Story	8	5	2000	2000	Gable	CompShg		
## 6	1.5Fin	5	5	1993	1995	Gable	CompShg		
##	Exterior1st	Exterior2nd	MasVnrType	MasVnrArea	ExterQual	ExterCond	Foundation		
## 1	VinylSd	VinylSd	BrkFace	196	Gd	TA	PConc		
## 2	MetalSd	MetalSd	None	0	TA	TA	CBlock		
## 3	VinylSd	VinylSd	BrkFace	162	Gd	TA	PConc		
## 4	Wd Sdng	Wd Shng	None	0	TA	TA	BrkTil		
## 5	VinylSd	VinylSd	BrkFace	350	Gd	TA	PConc		
## 6	VinylSd	VinylSd	None	0	TA	TA	Wood		
##	BsmtQual	BsmtCond	BsmtExposure	BsmtFinType1	BsmtFinSF1	BsmtFinType2			
## 1	Gd	TA	No	GLQ	706	Unf			
## 2	Gd	TA	Gd	ALQ	978	Unf			
## 3	Gd	TA	Mn	GLQ	486	Unf			
## 4	TA	Gd	No	ALQ	216	Unf			
## 5	Gd	TA	Av	GLQ	655	Unf			
## 6	Gd	TA	No	GLQ	732	Unf			
##	BsmtFinSF2	BsmtUnfSF	TotalBsmtSF	Heating	HeatingQC	CentralAir	Electrical		
## 1	0	150	856	GasA	Ex	Y	SBrkr		
## 2	0	284	1262	GasA	Ex	Y	SBrkr		
## 3	0	434	920	GasA	Ex	Y	SBrkr		
## 4	0	540	756	GasA	Gd	Y	SBrkr		
## 5	0	490	1145	GasA	Ex	Y	SBrkr		
## 6	0	64	796	GasA	Ex	Y	SBrkr		
##	X1stFlrSF	X2ndFlrSF	LowQualFinSF	GrLivArea	BsmtFullBath	BsmtHalfBath	FullBath		
## 1	856	854	0	1710	1	0	2		
## 2	1262	0	0	1262	0	1	2		
## 3	920	866	0	1786	1	0	2		
## 4	961	756	0	1717	1	0	1		
## 5	1145	1053	0	2198	1	0	2		
## 6	796	566	0	1362	1	0	1		
##	HalfBath	BedroomAbvGr	KitchenAbvGr	KitchenQual	TotRmsAbvGrd	Functional			
## 1	1	3	1	Gd	8	Typ			
## 2	0	3	1	TA	6	Typ			

```

## 3      1      3      1      Gd      6      Typ
## 4      0      3      1      Gd      7      Typ
## 5      1      4      1      Gd      9      Typ
## 6      1      1      1      TA      5      Typ
##  Fireplaces FireplaceQu GarageType GarageYrBlt GarageFinish GarageCars
## 1      0      <NA> Attchd      2003      RFn      2
## 2      1      TA      Attchd      1976      RFn      2
## 3      1      TA      Attchd      2001      RFn      2
## 4      1      Gd      Detchd      1998      Unf      3
## 5      1      TA      Attchd      2000      RFn      3
## 6      0      <NA> Attchd      1993      Unf      2
##  GarageArea GarageQual GarageCond PavedDrive WoodDeckSF OpenPorchSF
## 1      548      TA      TA      Y      0      61
## 2      460      TA      TA      Y      298      0
## 3      608      TA      TA      Y      0      42
## 4      642      TA      TA      Y      0      35
## 5      836      TA      TA      Y      192      84
## 6      480      TA      TA      Y      40      30
##  EnclosedPorch X3SsnPorch ScreenPorch PoolArea PoolQC Fence MiscFeature
## 1      0      0      0      0      <NA> <NA> <NA>
## 2      0      0      0      0      <NA> <NA> <NA>
## 3      0      0      0      0      <NA> <NA> <NA>
## 4      272      0      0      0      <NA> <NA> <NA>
## 5      0      0      0      0      <NA> <NA> <NA>
## 6      0      320      0      0      <NA> MnPrv Shed
##  MiscVal MoSold YrSold SaleType SaleCondition SalePrice economico medio caro
## 1      0      2      2008      WD      Normal      208500      1      0      0
## 2      0      5      2007      WD      Normal      181500      1      0      0
## 3      0      9      2008      WD      Normal      223500      1      0      0
## 4      0      2      2006      WD      Abnorml      140000      1      0      0
## 5      0      12      2008      WD      Normal      250000      1      0      0
## 6      700      10      2009      WD      Normal      143000      1      0      0

```

Se observa que utilizamos para la variable objetivo tres categorías, economico, caro y medio donde primero calculamos el valor de la casa máxima y el mínimo valor de la casa donde luego lo restamos el valor máximo con el mínimo y luego dividirlo entre tres para así diferenciar que casa debería de pertenecer en esas tres variables.

Solo se elimina la variable ID, Exterior2ndCBlock, GrLivArea, TotalBsmtSF porque no es importante en estos análisis y evitar el análisis de más variables.

```
summary(modelo_todas_cv)
```

```

##
## Call:
## NULL
##
## Coefficients: (3 not defined because of singularities)
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    2.657e+01  1.654e+07      0      1
## MSSubClass      6.661e-15  1.301e+03      0      1
## MSZoningFV     -1.533e-13  1.889e+05      0      1
## MSZoningRH      4.699e-13  1.872e+05      0      1
## MSZoningRL      3.488e-13  1.608e+05      0      1
## MSZoningRM      4.195e-13  1.505e+05      0      1
## LotFrontage    -1.545e-14  6.903e+02      0      1
## LotArea        -1.213e-16  1.749e+00      0      1
## StreetPave     -3.577e-12  1.904e+05      0      1
## AlleyPave       1.715e-12  7.469e+04      0      1
## LotShapeIR2    -4.465e-13  6.640e+04      0      1
## LotShapeIR3     1.419e-12  1.398e+05      0      1
## LotShapeReg    -6.592e-13  2.516e+04      0      1
## LandContourHLS -2.003e-12  8.051e+04      0      1
## LandContourLow  1.793e-12  1.001e+05      0      1
## LandContourLvl  2.327e-14  5.794e+04      0      1
## UtilitiesNoSeWa 8.389e-12  4.145e+05      0      1
## LotConfigCulDSac -1.878e-12  5.206e+04      0      1
## LotConfigFR2   -2.475e-13  6.332e+04      0      1
## LotConfigFR3   -2.153e-12  1.976e+05      0      1
## LotConfigInside -2.748e-13  2.814e+04      0      1
## LandSlopeMod    5.262e-13  6.254e+04      0      1
## LandSlopeSev    7.730e-12  1.803e+05      0      1
## NeighborhoodBlueste 8.778e-13  3.032e+05      0      1
## NeighborhoodBrDale -2.786e-15  1.725e+05      0      1
## NeighborhoodBrkSide 1.314e-12  1.489e+05      0      1
## NeighborhoodClearCr 2.268e-12  1.446e+05      0      1
## NeighborhoodCollgCr 2.118e-12  1.141e+05      0      1
## NeighborhoodCrawfor 2.427e-12  1.346e+05      0      1
## NeighborhoodEdwards 3.516e-12  1.261e+05      0      1
## NeighborhoodGilbert 1.773e-12  1.207e+05      0      1
## NeighborhoodIDOTRR 5.281e-12  1.688e+05      0      1
## NeighborhoodMeadowV 1.164e-12  1.763e+05      0      1
## NeighborhoodMitchel 1.662e-12  1.286e+05      0      1
## NeighborhoodNAMES 1.807e-12  1.232e+05      0      1
## NeighborhoodNoRidge 5.046e-12  1.339e+05      0      1
## NeighborhoodNPkVill -5.475e-13  2.213e+05      0      1
## NeighborhoodNridgHt 1.322e-12  1.193e+05      0      1
## NeighborhoodNWAMES 2.114e-12  1.262e+05      0      1
## NeighborhoodOldTown 5.554e-12  1.510e+05      0      1
## NeighborhoodSawyer 2.579e-12  1.278e+05      0      1
## NeighborhoodSawyerW 1.988e-12  1.224e+05      0      1
## NeighborhoodSomerst 1.683e-12  1.418e+05      0      1
## NeighborhoodStoneBr -2.015e-12  1.319e+05      0      1
## NeighborhoodSWISU 4.673e-12  1.525e+05      0      1
## NeighborhoodTimber 1.144e-12  1.275e+05      0      1

```

## NeighborhoodVeenker	3.504e-13	1.653e+05	0	1
## Condition1Feedr	4.815e-12	7.859e+04	0	1
## Condition1Norm	4.107e-12	6.596e+04	0	1
## Condition1PosA	5.164e-12	1.571e+05	0	1
## Condition1PosN	4.473e-12	1.169e+05	0	1
## Condition1RR Ae	4.870e-12	1.427e+05	0	1
## Condition1RR An	3.474e-12	1.090e+05	0	1
## Condition1RR Ne	5.202e-12	2.748e+05	0	1
## Condition1RR Nn	-5.933e-13	2.028e+05	0	1
## Condition2Feedr	2.546e-12	3.681e+05	0	1
## Condition2Norm	-4.530e-12	3.183e+05	0	1
## Condition2PosA	7.785e-12	5.831e+05	0	1
## Condition2PosN	6.791e-12	4.484e+05	0	1
## Condition2RR Ae	1.410e-11	8.951e+05	0	1
## Condition2RR An	-4.819e-12	4.955e+05	0	1
## Condition2RR Nn	-3.474e-12	4.250e+05	0	1
## BldgType2fmCon	-2.640e-12	1.965e+05	0	1
## BldgTypeDuplex	-1.713e-12	1.157e+05	0	1
## BldgTypeTwnhs	-4.017e-13	1.580e+05	0	1
## BldgTypeTwnhsE	3.541e-13	1.422e+05	0	1
## HouseStyle1.5Unf	-5.766e-13	1.243e+05	0	1
## HouseStyle1Story	-1.336e-12	6.860e+04	0	1
## HouseStyle2.5Fin	1.080e-11	1.942e+05	0	1
## HouseStyle2.5Unf	-2.510e-12	1.453e+05	0	1
## HouseStyle2Story	-1.723e-12	5.502e+04	0	1
## HouseStyleSFoyer	-8.396e-13	9.788e+04	0	1
## HouseStyleSLvl	-9.990e-13	8.699e+04	0	1
## OverallQual	1.036e-13	1.631e+04	0	1
## OverallCond	5.208e-13	1.399e+04	0	1
## YearBuilt	-4.551e-15	1.208e+03	0	1
## YearRemodAdd	-3.681e-14	8.758e+02	0	1
## RoofStyleGable	4.064e-12	2.901e+05	0	1
## RoofStyleGambrel	4.350e-12	3.173e+05	0	1
## RoofStyleHip	3.940e-12	2.913e+05	0	1
## RoofStyleMansard	3.202e-12	3.369e+05	0	1
## RoofStyleShed	-4.901e-12	5.456e+05	0	1
## RoofMatlCompShg	-3.391e-11	6.426e+05	0	1
## RoofMatlMembran	-3.073e-11	8.608e+05	0	1
## RoofMatlMetal	-3.811e-11	8.403e+05	0	1
## RoofMatlRoll	-2.966e-11	7.506e+05	0	1
## `RoofMatlTar&Grv`	-2.907e-11	7.013e+05	0	1
## RoofMatlWdShake	-3.427e-11	6.880e+05	0	1
## RoofMatlWdShngl	-3.471e-11	6.686e+05	0	1
## Exterior1stAsphShn	-5.949e-12	5.153e+05	0	1
## Exterior1stBrkComm	1.691e-12	4.351e+05	0	1
## Exterior1stBrkFace	1.219e-12	1.994e+05	0	1
## Exterior1stCBlock	8.826e-12	4.274e+05	0	1
## Exterior1stCemntBd	-3.590e-11	3.040e+05	0	1
## Exterior1stHdBoard	-5.367e-13	2.023e+05	0	1
## Exterior1stImStucc	-1.644e-11	4.436e+05	0	1
## Exterior1stMetalSd	4.740e-13	2.288e+05	0	1
## Exterior1stPlywood	-7.452e-13	1.998e+05	0	1

## Exterior1stStone	6.152e-12	3.767e+05	0	1
## Exterior1stStucco	-4.716e-12	2.215e+05	0	1
## Exterior1stVinylSd	2.108e-12	2.081e+05	0	1
## `Exterior1stWd Sdng`	4.409e-12	1.931e+05	0	1
## Exterior1stWdShng	1.537e-12	2.092e+05	0	1
## Exterior2ndAsphShn	4.137e-12	3.485e+05	0	1
## `Exterior2ndBrk Cmn`	6.902e-13	3.151e+05	0	1
## Exterior2ndBrkFace	-1.247e-12	2.060e+05	0	1
## Exterior2ndCBlock	NA	NA	NA	NA
## Exterior2ndCmentBd	3.728e-11	3.001e+05	0	1
## Exterior2ndHdBoard	1.716e-12	1.945e+05	0	1
## Exterior2ndImStucc	2.061e-11	2.252e+05	0	1
## Exterior2ndMetalSd	1.114e-12	2.229e+05	0	1
## Exterior2ndOther	4.558e-12	4.270e+05	0	1
## Exterior2ndPlywood	1.553e-12	1.889e+05	0	1
## Exterior2ndStone	-1.339e-12	2.668e+05	0	1
## Exterior2ndStucco	-5.230e-12	2.143e+05	0	1
## Exterior2ndVinylSd	-6.318e-13	2.010e+05	0	1
## `Exterior2ndWd Sdng`	-2.578e-12	1.870e+05	0	1
## `Exterior2ndWd Shng`	-1.936e-12	1.955e+05	0	1
## MasVnrTypeBrkFace	-2.249e-12	1.076e+05	0	1
## MasVnrTypeNone	-2.203e-12	1.086e+05	0	1
## MasVnrTypeStone	-2.595e-12	1.138e+05	0	1
## MasVnrArea	-3.952e-15	9.123e+01	0	1
## ExterQualFa	8.387e-12	1.741e+05	0	1
## ExterQualGd	7.456e-12	7.594e+04	0	1
## ExterQualTA	9.024e-12	8.380e+04	0	1
## ExterCondFa	5.801e-12	2.824e+05	0	1
## ExterCondGd	1.060e-12	2.699e+05	0	1
## ExterCondPo	9.093e-12	4.948e+05	0	1
## ExterCondTA	2.250e-12	2.693e+05	0	1
## FoundationCBlock	-3.620e-13	4.991e+04	0	1
## FoundationPConc	-2.265e-12	5.381e+04	0	1
## FoundationSlab	8.230e-13	1.233e+05	0	1
## FoundationStone	2.471e-12	1.769e+05	0	1
## FoundationWood	-2.084e-12	2.325e+05	0	1
## BsmtQualFa	-2.740e-12	9.951e+04	0	1
## BsmtQualGd	-3.467e-12	5.389e+04	0	1
## BsmtQualTA	-3.393e-12	6.598e+04	0	1
## BsmtCondGd	-1.943e-12	8.297e+04	0	1
## BsmtCondPo	-1.524e-11	4.703e+05	0	1
## BsmtCondTA	-1.665e-12	6.669e+04	0	1
## BsmtExposureGd	-2.608e-12	4.779e+04	0	1
## BsmtExposureMn	3.545e-14	4.756e+04	0	1
## BsmtExposureNo	-1.427e-13	3.429e+04	0	1
## BsmtFinType1BLQ	5.532e-13	4.386e+04	0	1
## BsmtFinType1GLQ	4.038e-13	3.971e+04	0	1
## BsmtFinType1LwQ	9.449e-13	5.885e+04	0	1
## BsmtFinType1Rec	1.548e-12	4.710e+04	0	1
## BsmtFinType1Unf	4.131e-13	4.550e+04	0	1
## BsmtFinSF1	6.287e-17	7.473e+01	0	1
## BsmtFinType2BLQ	3.962e-13	1.187e+05	0	1

## BsmtFinType2GLQ	-1.447e-12	1.474e+05	0	1
## BsmtFinType2LwQ	-4.191e-12	1.162e+05	0	1
## BsmtFinType2Rec	-4.805e-13	1.118e+05	0	1
## BsmtFinType2Unf	1.102e-13	1.183e+05	0	1
## BsmtFinSF2	2.161e-15	1.343e+02	0	1
## BsmtUnfSF	-1.396e-16	6.293e+01	0	1
## TotalBsmtSF	NA	NA	NA	NA
## HeatingGasA	2.104e-12	3.867e+05	0	1
## HeatingGasW	4.472e-12	4.006e+05	0	1
## HeatingGrav	5.751e-13	4.259e+05	0	1
## HeatingOthW	3.451e-12	4.850e+05	0	1
## HeatingWall	8.756e-12	4.500e+05	0	1
## HeatingQCFA	-1.317e-12	7.416e+04	0	1
## HeatingQCGd	-6.874e-13	3.253e+04	0	1
## HeatingQCPo	-1.435e-12	4.180e+05	0	1
## HeatingQCTA	1.930e-13	3.258e+04	0	1
## CentralAirY	-1.391e-12	6.081e+04	0	1
## ElectricalFuseF	1.177e-13	9.041e+04	0	1
## ElectricalFuseP	-1.714e-12	2.899e+05	0	1
## ElectricalMix	1.564e-12	7.000e+05	0	1
## ElectricalSBrkr	-2.232e-13	4.626e+04	0	1
## X1stFlrSF	-2.889e-16	8.657e+01	0	1
## X2ndFlrSF	-1.529e-15	9.389e+01	0	1
## LowQualFinSF	-4.523e-14	2.954e+02	0	1
## GrLivArea	NA	NA	NA	NA
## BsmtFullBath	-3.463e-13	3.110e+04	0	1
## BsmtHalfBath	-2.140e-12	4.758e+04	0	1
## FullBath	-1.709e-13	3.469e+04	0	1
## HalfBath	9.954e-13	3.297e+04	0	1
## BedroomAbvGr	3.263e-13	2.152e+04	0	1
## KitchenAbvGr	1.719e-12	8.914e+04	0	1
## KitchenQualFa	-2.590e-12	9.784e+04	0	1
## KitchenQualGd	-1.837e-12	5.585e+04	0	1
## KitchenQualTA	-1.554e-12	6.257e+04	0	1
## TotRmsAbvGrd	-3.200e-13	1.504e+04	0	1
## FunctionalMaj2	6.646e-12	2.265e+05	0	1
## FunctionalMin1	-1.694e-12	1.354e+05	0	1
## FunctionalMin2	-2.800e-12	1.357e+05	0	1
## FunctionalMod	-1.147e-11	1.661e+05	0	1
## FunctionalSev	2.264e-14	4.649e+05	0	1
## FunctionalTyp	-2.455e-12	1.176e+05	0	1
## Fireplaces	3.104e-13	2.401e+04	0	1
## FireplaceQuFa	9.691e-12	1.083e+05	0	1
## FireplaceQuGd	9.266e-12	8.392e+04	0	1
## FireplaceQuPo	9.801e-12	1.233e+05	0	1
## FireplaceQuTA	9.822e-12	8.695e+04	0	1
## GarageTypeAttchd	-2.049e-12	1.735e+05	0	1
## GarageTypeBasment	-1.629e-12	2.004e+05	0	1
## GarageTypeBuiltIn	-3.053e-12	1.806e+05	0	1
## GarageTypeCarPort	-3.931e-12	2.305e+05	0	1
## GarageTypeDetchd	-2.283e-12	1.729e+05	0	1
## GarageYrBlt	5.361e-14	8.923e+02	0	1

## GarageFinishRFn	2.004e-13	3.098e+04	0	1
## GarageFinishUnf	3.763e-14	3.773e+04	0	1
## GarageCars	1.250e-12	3.453e+04	0	1
## GarageArea	-6.820e-15	1.224e+02	0	1
## GarageQualFa	-9.740e-12	4.832e+05	0	1
## GarageQualGd	-1.227e-11	4.941e+05	0	1
## GarageQualPo	-1.076e-11	6.125e+05	0	1
## GarageQualTA	-1.024e-11	4.782e+05	0	1
## GarageCondFa	1.361e-11	5.528e+05	0	1
## GarageCondGd	1.559e-11	5.731e+05	0	1
## GarageCondPo	1.534e-11	5.924e+05	0	1
## GarageCondTA	1.328e-11	5.482e+05	0	1
## PavedDriveP	2.144e-12	8.597e+04	0	1
## PavedDriveY	2.028e-13	5.379e+04	0	1
## WoodDeckSF	-1.217e-15	9.257e+01	0	1
## OpenPorchSF	-5.785e-15	1.806e+02	0	1
## EnclosedPorch	-2.526e-14	1.961e+02	0	1
## X3SsnPorch	-2.427e-16	3.520e+02	0	1
## ScreenPorch	-1.085e-14	1.973e+02	0	1
## PoolArea	9.296e-15	4.636e+02	0	1
## PoolQCFa	-2.021e-10	5.004e+05	0	1
## PoolQCGd	-2.018e-10	4.034e+05	0	1
## FenceGdWo	3.809e-12	7.735e+04	0	1
## FenceMnPrv	3.733e-12	5.704e+04	0	1
## FenceMnWw	4.546e-12	1.292e+05	0	1
## MiscFeatureOthr	2.505e-11	1.006e+06	0	1
## MiscFeatureShed	1.930e-11	1.038e+06	0	1
## MiscFeatureTenC	1.812e-11	1.070e+06	0	1
## MiscVal	1.252e-15	6.304e+01	0	1
## MoSold	8.678e-14	3.858e+03	0	1
## YrSold	-1.270e-14	8.101e+03	0	1
## SaleTypeCon	-1.459e-12	2.768e+05	0	1
## SaleTypeConLD	1.039e-12	1.526e+05	0	1
## SaleTypeConLI	-2.007e-12	1.813e+05	0	1
## SaleTypeConLw	-7.088e-13	1.908e+05	0	1
## SaleTypeCWD	-1.570e-12	2.027e+05	0	1
## SaleTypeNew	3.616e-12	2.427e+05	0	1
## SaleTypeOth	3.002e-13	2.255e+05	0	1
## SaleTypeWD	-9.568e-13	6.570e+04	0	1
## SaleConditionAdjLand	-1.266e-12	2.276e+05	0	1
## SaleConditionAlloca	-7.232e-13	1.364e+05	0	1
## SaleConditionFamily	5.142e-13	9.558e+04	0	1
## SaleConditionNormal	2.484e-13	4.570e+04	0	1
## SaleConditionPartial	-4.573e-12	2.333e+05	0	1
## SalePrice	5.242e-17	5.647e-01	0	1
## economico	-5.313e+01	2.113e+05	0	1
## medio	-5.313e+01	1.853e+05	0	1
##				
## (Dispersion parameter for binomial family taken to be 1)				
##				
## Null deviance: 9.9264e+01 on 1459 degrees of freedom				
## Residual deviance: 8.4703e-09 on 1214 degrees of freedom				

```
## AIC: 492
##
## Number of Fisher Scoring iterations: 25
```

El modelo de regresión logística con validación cruzada mostró que ninguna de las variables es significativa, ya que los valores de z son 0 y los valores de $\Pr(>|z|)$ son 1, lo que indica que las variables no están contribuyendo de manera relevante. Esto podría ser causado por la multicolinealidad entre las variables, debido a que muchas de ellas están altamente correlacionadas, lo que afecta la estimación de los coeficientes. Otra posible causa es el exceso de variables en relación con el número de observaciones, lo que podría llevar a un sobreajuste lo cual es más probable con los modelos realizadas con anterioridad.

```
## Call:
## nnet::multinom(formula = .outcome ~ ., data = dat, decay = param$decay,
##   trace = FALSE)
##
## Coefficients:
##      (Intercept)  MSSubClass MSZoningFV MSZoningRH  MSZoningRL MSZoningRM
## economico -0.01714893  0.001101410  0.1530568  0.2957217 -0.05689274 -0.2604506
## medio      0.01312096 -0.006694639  0.3162988 -0.4647028  0.82131934  1.1182811
##      LotFrontage      LotArea StreetPave  AlleyPave LotShapeIR2
## economico -0.01299268  0.0001491241 -0.5441204  0.20742026 -0.2246062
## medio      -0.01687481  0.0001036350  1.9050850 -0.02189445 -0.5503255
##      LotShapeIR3 LotShapeReg LandContourHLS LandContourLow LandContourLvl
## economico  1.1490900 -0.03284831 -0.28870227  0.5214846  0.3077974
## medio      0.9912686 -0.39390725 -0.06627513  0.6278728  0.1867241
##      UtilitiesNoSewa LotConfigCulDSac LotConfigFR2 LotConfigFR3
## economico -0.3790245 -0.4303433 -0.29089932 -0.4764919
## medio      0.7074540 -0.2424822 -0.09335567  0.1178051
##      LotConfigInside LandSlopeMod LandSlopeSev NeighborhoodBlueste
## economico -0.2662634 -0.1026328  0.4810144  0.1771096
## medio      -0.3429775  0.1027424 -0.2696511 -0.3309488
##      NeighborhoodBrDale NeighborhoodBrkSide NeighborhoodClearCr
## economico  0.09232272  0.4122681  0.8579764
## medio      -0.05203416  0.2004710  0.3427614
##      NeighborhoodCollgCr NeighborhoodCrawfor NeighborhoodEdwards
## economico -0.6141003  0.8707533  0.07641929
## medio      0.3972050  0.1707178 -0.09984035
##      NeighborhoodGilbert NeighborhoodIDOTRR NeighborhoodMeadowV
## economico -0.20657514 -0.2520061 -0.34360850
## medio      -0.04037338 -0.4225894  0.07932858
##      NeighborhoodMitchel NeighborhoodNames NeighborhoodNoRidge
## economico -0.01416389  0.0207002 -0.3561949
## medio      -0.10715137 -0.1621490  0.3524228
##      NeighborhoodNPkVill NeighborhoodNridgHt NeighborhoodNWames
## economico -0.03353481 -0.090126027 -0.1220062
## medio      0.56598072  0.006903909  0.1874474
##      NeighborhoodOldTown NeighborhoodSawyer NeighborhoodSawyerW
## economico  0.07742748  0.1042675 -0.3875433
## medio      -0.27604055  0.7234797 -0.3815222
##      NeighborhoodSomerst NeighborhoodStoneBr NeighborhoodSWISU
## economico  0.1984696 -0.88008644  0.30924458
## medio      -0.2794237  0.03543834 -0.07452418
##      NeighborhoodTimber NeighborhoodVeenker Condition1Feedr Condition1Norm
## economico  0.4773896  0.54821107  0.3191159  0.05091740
## medio      0.4860941  0.05034178 -0.1661514  0.03322942
##      Condition1PosA Condition1PosN Condition1RR Ae Condition1RRAn
## economico  0.4552429  0.2413831  0.4151138 -0.07765070
## medio      0.1544174  0.3099631 -0.4416228 -0.05919175
##      Condition1RRNe Condition1RRNn Condition2Feedr Condition2Norm
## economico -0.52754577 -0.3865589  0.601042  0.4077817
## medio      0.01462837 -0.4298720 -1.350585  0.8336800
##      Condition2PosA Condition2PosN Condition2RR Ae Condition2RRAn
## economico -0.4121201 -0.53711508 -0.1980976 -0.20494656
```

## medio	0.1712187	0.07148083	0.4430674	0.08656668
##	Condition2RRNn	BldgType2fmCon	BldgTypeDuplex	BldgTypeTwnhs
## economico	0.10872284	-0.2213977	-0.01106503	-0.3203615
## medio	-0.07571559	0.2348846	-0.27582705	0.6400671
##	BldgTypeTwnhsE	HouseStyle1.5Unf	HouseStyle1Story	HouseStyle2.5Fin
## economico	-0.12936422	0.3139577	-0.03307198	0.2366385
## medio	-0.06738761	-0.5213995	-0.38632283	0.4281893
##	HouseStyle2.5Unf	HouseStyle2Story	HouseStyleSFoyer	HouseStyleSLvl
## economico	-0.3161256	-0.008236034	0.14599529	-0.3603264
## medio	-0.3267774	0.356919350	-0.07447617	-0.2552559
##	OverallQual	OverallCond	YearBuilt	YearRemodAdd
## economico	0.4098179	0.3226525	0.01518136	0.005503871
## medio	0.1220051	0.7728217	0.03064422	-0.016057561
##	RoofStyleGambrel	RoofStyleHip	RoofStyleMansard	RoofStyleShed
## economico	0.053505473	-0.4552475	-0.001746087	-0.2011856
## medio	0.002144258	-1.0616212	0.600209922	0.4568513
##	RoofMatlCompShg	RoofMatlMembran	RoofMatlMetal	RoofMatlRoll
## economico	0.03653475	0.32026835	0.2974809	0.2993835
## medio	0.43899540	-0.01349971	0.1097386	0.1285226
##	`RoofMatlTar&Grv`	RoofMatlWdShake	RoofMatlWdShngl	Exterior1stAsphShn
## economico	-0.2127485	-0.19032503	-0.4767200	-0.3179992
## medio	0.3525885	-0.07901624	-0.4340175	0.1326490
##	Exterior1stBrkComm	Exterior1stBrkFace	Exterior1stCBlock	
## economico	-0.16359160	0.48116447	0.06121009	
## medio	0.07876888	-0.01821254	-0.09771173	
##	Exterior1stCemntBd	Exterior1stHdBoard	Exterior1stImStucc	
## economico	-1.5019220	-0.1172292	0.1150858	
## medio	-0.7030497	-0.1473618	-0.1137626	
##	Exterior1stMetalSd	Exterior1stPlywood	Exterior1stStone	
## economico	0.4450422	0.7053514	0.1203959	
## medio	-0.1514494	-0.1794610	-0.3794658	
##	Exterior1stStucco	Exterior1stVinylSd	`Exterior1stWd Sdng`	
## economico	-0.5637144	0.7217669	0.2131426	
## medio	0.7357022	-0.2902308	-0.1138478	
##	Exterior1stWdShing	Exterior2ndAsphShn	`Exterior2ndBrk Cmn`	
## economico	0.52076672	0.3276143	0.06730893	
## medio	0.01953439	-0.2802776	0.31734865	
##	Exterior2ndBrkFace	Exterior2ndCBlock	Exterior2ndCmentBd	
## economico	-0.2195023	0.06121009	0.84910659	
## medio	0.1078082	-0.09771173	0.02124405	
##	Exterior2ndHdBoard	Exterior2ndImStucc	Exterior2ndMetalSd	
## economico	-0.7065934	-0.07080637	-0.2893852	
## medio	-0.1985426	-0.09238393	-0.1014440	
##	Exterior2ndOther	Exterior2ndPlywood	Exterior2ndStone	
## economico	0.0537975	-0.1877785	-0.3195776	
## medio	-0.1672481	-0.2087765	0.3462856	
##	Exterior2ndStucco	Exterior2ndVinylSd	`Exterior2ndWd Sdng`	
## economico	0.7306478	-0.11743038	0.004325423	
## medio	-0.1457104	-0.01608508	0.038003628	
##	`Exterior2ndWd Shng`	MasVnrTypeBrkFace	MasVnrTypeNone	MasVnrTypeStone
## economico	-0.6456399	-0.01680262	0.33396639	-0.06364656
## medio	-0.1597615	-0.28403460	0.09018723	0.16693987

```

##      MasVnrArea  ExterQualFa  ExterQualGd  ExterQualTA  ExterCondFa
## economico 0.001583670 -0.07678328  0.4148993 -0.3057784  0.2202689
## medio    0.001607949 -0.87318144  -0.9032618 -1.1099690 -0.1694694
##      ExterCondGd  ExterCondPo  ExterCondTA  FoundationCBlock  FoundationPConc
## economico 0.36855995  0.04833628  0.1493948  0.1714171  0.4462299
## medio    -0.03047594  0.14765172  -0.1995359  0.2197555  0.7311633
##      FoundationSlab  FoundationStone  FoundationWood  BsmtQualFa  BsmtQualGd
## economico -0.6480286  1.438981  0.6807202 -0.4766127  0.1333541
## medio    -0.1745766  0.651968  -0.6031166  0.1479070  0.6127214
##      BsmtQualTA  BsmtCondGd  BsmtCondPo  BsmtCondTA  BsmtExposureGd
## economico 0.1831356  0.4493871  0.3855699 -0.3305763  -0.4371681
## medio    -0.2544266 -0.4158386 -0.7869508  0.4019760  -0.4823639
##      BsmtExposureMn  BsmtExposureNo  BsmtFinType1BLQ  BsmtFinType1GLQ
## economico 0.3316429  0.2872012  0.03156766  0.1725523
## medio    -0.2854998  -0.3415710  0.12686212  0.4102017
##      BsmtFinType1LwQ  BsmtFinType1Rec  BsmtFinType1Unf  BsmtFinSF1
## economico 0.08289960  -0.1537751  -0.1133463 -0.0003044563
## medio    -0.06443489  -0.2297761  0.1163508 -0.0004281523
##      BsmtFinType2BLQ  BsmtFinType2GLQ  BsmtFinType2LwQ  BsmtFinType2Rec
## economico 0.2830420  0.01488595  0.4146735  0.1552275
## medio    -0.1152676  -0.04784530  0.2965416  0.5653399
##      BsmtFinType2Unf  BsmtFinSF2  BsmtUnfSF  TotalBsmtSF  HeatingGasA
## economico 0.2371898 -0.0001721706 -0.0003553337 -0.0008319681 -0.7506293
## medio    0.5097313  0.0006826186 -0.0004627562 -0.0002082921 -0.3838489
##      HeatingGasW  HeatingGrav  HeatingOthW  HeatingWall  HeatingQCFA
## economico 0.5109077  -0.2522827  0.06104162  0.11413934  0.06258231
## medio    0.8908967  -0.2538473  -0.06823086 -0.09284914 -0.05108432
##      HeatingQCgd  HeatingQCpo  HeatingQCTA  CentralAirY  ElectricalFuseF
## economico 0.1055135  -0.1413208 -0.12858786  0.1073664  -0.04027725
## medio    -0.1480589  0.1183587 -0.08465793  0.4735902  0.28771567
##      ElectricalFuseP  ElectricalMix  ElectricalSBrkr  X1stFlrSF
## economico 0.09306355  0.2829388  0.4308974  9.467992e-05
## medio    -0.08511168  -0.3774419  0.6376577 -2.985019e-04
##      X2ndFlrSF  LowQualFinSF  GrLivArea  BsmtFullBath  BsmtHalfBath
## economico -0.0009751435  0.003073956  0.002193534  0.5262366  0.04053264
## medio    -0.0018323491  0.003778733  0.001647851  -0.2288404  -0.05947129
##      FullBath  HalfBath  BedroomAbvGr  KitchenAbvGr  KitchenQualFa
## economico 0.1930876  0.03125897  0.8890714  0.1752478  -0.130697
## medio    0.4375040  0.02837515  -0.8042966  0.2776410  -0.288850
##      KitchenQualGd  KitchenQualTA  TotRmsAbvGrd  FunctionalMaj2
## economico 0.23097437  0.2996015  -0.6577827  1.8515709
## medio    0.08221661  -0.3058388  0.3079607  -0.9223682
##      FunctionalMin1  FunctionalMin2  FunctionalMod  FunctionalSev
## economico -0.2850374  0.01360182  -1.3558075  0.3758713
## medio    0.3242594  0.58516888  -0.2171951  -0.7033537
##      FunctionalTyp  Fireplaces  FireplaceQuFa  FireplaceQuGd  FireplaceQuPo
## economico -0.2049661  0.4809739  -0.4609171  -1.0415242  -0.7615189
## medio    0.6885652  0.5366603  -0.3631750  -0.4783431  -0.2812913
##      FireplaceQuTA  GarageTypeAttchd  GarageTypeBasment  GarageTypeBuiltIn
## economico -0.8728485  0.07246153  0.1371366  0.04449576
## medio    -0.7370135  -0.39216198  0.7122471  0.14670151
##      GarageTypeCarPort  GarageTypeDetchd  GarageYrBlt  GarageFinishRFn

```

```

## economico      -0.1610016      0.006406593  0.0002712129      0.02656806
## medio          -0.7544620      -0.184441077 -0.0019906569      -0.17405334
##      GarageFinishUnf GarageCars      GarageArea GarageQualFa GarageQualGd
## economico      0.33036404  0.18366247 -0.0003501396  -0.36424658  -0.2193659
## medio          0.02684333 -0.02114018  0.0001952360  0.06002847  0.2364312
##      GarageQualPo GarageQualTA GarageCondFa GarageCondGd GarageCondPo
## economico      1.0945677  -0.67749935  -0.3563443  -0.4375111  0.7355742
## medio          -0.8563868  -0.03859975  0.5380417  0.1596827  -0.7560980
##      GarageCondTA PavedDriveP PavedDriveY      WoodDeckSF OpenPorchSF
## economico      -0.2895975  -0.7865759  -0.3764457  0.0003986864 -0.001866176
## medio          0.3487040  -0.5764269  -0.4030238  -0.0013835478 -0.002585929
##      EnclosedPorch X3SsnPorch ScreenPorch      PoolArea PoolQCFa
## economico      0.003546126 0.012549937 -0.001965906 0.003059456 -0.02054309
## medio          0.007759641 0.007561441 0.001412170 0.003931773 0.37359978
##      PoolQCGd FenceGdWo FenceMnPrv FenceMnWw MiscFeature0thr
## economico      -0.8259283 -0.3665405 -0.34011755 -0.05119467 0.02097904
## medio          -0.5891223 -0.2126394 0.05193031 -0.20175737 0.07356371
##      MiscFeatureShed MiscFeatureTenC      MiscVal      MoSold
## economico      -0.1090484  -0.04283081 0.0005769711 0.0008385918
## medio          -0.7836459  0.84486441 -0.0005527146 -0.0244102735
##      YrSold SaleTypeCon SaleTypeConLD SaleTypeConLI SaleTypeConLw
## economico      -0.01980608 0.4177099 0.5036697 0.2127623 -0.02612279
## medio          -0.01437454 -0.3320231 0.1212872 0.9382414 0.24388227
##      SaleTypeCWD SaleTypeNew SaleTypeOth SaleTypeWD SaleConditionAdjLand
## economico      0.4006678 0.2053054 0.3017478 -0.09414143 -0.5831784
## medio          0.3024123 0.1679612 -0.1213645 0.31955287 0.6171371
##      SaleConditionAlloca SaleConditionFamily SaleConditionNormal
## economico      0.5008348 0.02074259 -0.03052446
## medio          -1.3604822 0.32027298 0.34547539
##      SaleConditionPartial SalePrice economico medio caro
## economico      -0.1129704 -2.554907e-05 12.480314 -4.934031 -7.563431
## medio          0.3138578 -2.006937e-05 -1.498526 12.915822 -11.404175
##
## Std. Errors:
##      (Intercept) MSSubClass MSZoningFV MSZoningRH MSZoningRL
## economico      8.006962e-06 0.1557881 8.744219e-05 2.949224e-05 0.0005235057
## medio          7.604434e-06 0.0984562 1.521733e-04 2.856644e-06 0.0004211864
##      MSZoningRM LotFrontage LotArea StreetPave AlleyPave
## economico      0.0004922520 0.03414598 0.003347460 4.542891e-05 6.294235e-05
## medio          0.0003514056 0.02703459 0.003343605 7.592089e-06 1.829386e-05
##      LotShapeIR2 LotShapeIR3 LotShapeReg LandContourHLS LandContourLow
## economico      1.056805e-04 1.707236e-05 0.0002734384 0.0001101008 4.635251e-05
## medio          9.914008e-05 1.661205e-05 0.0003658016 0.0001405420 2.793003e-05
##      LandContourLvl UtilitiesNoSeWa LotConfigCulDSac LotConfigFR2
## economico      0.0002503898 0.0002355811 0.0003112939 9.306908e-05
## medio          0.0001641265 0.0002624652 0.0003122132 1.173668e-04
##      LotConfigFR3 LotConfigInside LandSlopeMod LandSlopeSev
## economico      1.239420e-05 0.0002019750 0.0001255374 2.89539e-05
## medio          1.148346e-05 0.0002224911 0.0001575924 3.88484e-05
##      NeighborhoodBlueste NeighborhoodBrDale NeighborhoodBrkSide
## economico      2.455551e-06 1.246459e-04 1.892147e-04
## medio          1.634291e-06 7.567216e-05 8.284681e-05

```

```

##      NeighborhoodClearCr NeighborhoodCollgCr NeighborhoodCrawfor
## economico      8.718677e-05      0.0003883228      7.221292e-05
## medio          7.269436e-05      0.0002967880      1.404745e-04
##      NeighborhoodEdwards NeighborhoodGilbert NeighborhoodIDOTRR
## economico      1.020884e-04      0.0001357913      1.919710e-04
## medio          6.453488e-05      0.0001131858      7.639078e-05
##      NeighborhoodMeadowV NeighborhoodMitchel NeighborhoodNAmes
## economico      0.0003197793      5.568615e-05      0.0002894844
## medio          0.0002294299      3.531849e-05      0.0001526972
##      NeighborhoodNoRidge NeighborhoodNPkVill NeighborhoodNridgHt
## economico      8.691373e-05      4.645447e-05      0.0002374752
## medio          6.047711e-05      2.965833e-05      0.0002489233
##      NeighborhoodNWAmes NeighborhoodOldTown NeighborhoodSawyer
## economico      0.0001929931      0.0002923957      1.181406e-04
## medio          0.0001272635      0.0001917203      6.481616e-05
##      NeighborhoodSawyerW NeighborhoodSomerst NeighborhoodStoneBr
## economico      5.080920e-05      0.0001033116      0.00017405
## medio          4.387882e-05      0.0001726743      0.00018882
##      NeighborhoodSWISU NeighborhoodTimber NeighborhoodVeenker
## economico      2.150230e-05      0.0002282526      1.180398e-05
## medio          1.921472e-05      0.0002365110      1.397384e-05
##      Condition1Feedr Condition1Norm Condition1PosA Condition1PosN
## economico      7.873359e-05      0.0001244117      4.426290e-05      2.758248e-05
## medio          5.622046e-05      0.0001015582      7.150297e-05      4.187672e-05
##      Condition1RRAE Condition1RRAn Condition1RRNe Condition1RRNn
## economico      8.794045e-06      9.193757e-05      1.713039e-05      3.932509e-05
## medio          1.515519e-05      5.546320e-05      1.591023e-05      7.726112e-06
##      Condition2Feedr Condition2Norm Condition2PosA Condition2PosN
## economico      9.033212e-06      6.918377e-05      8.870379e-09      2.435800e-05
## medio          2.530654e-06      6.773125e-05      4.760552e-05      2.200284e-05
##      Condition2RRAE Condition2RRAn Condition2RRNn BldgType2fmCon
## economico      2.192364e-07      1.339291e-06      1.009610e-05      2.013319e-04
## medio          3.441648e-07      8.055687e-07      6.019827e-06      3.256161e-05
##      BldgTypeDuplex BldgTypeTwnhs BldgTypeTwnhsE HouseStyle1.5Unf
## economico      1.284857e-04      0.0004748924      0.0006239287      2.178026e-05
## medio          5.679441e-05      0.0003829690      0.0004158675      1.676275e-05
##      HouseStyle1Story HouseStyle2.5Fin HouseStyle2.5Unf HouseStyle2Story
## economico      0.0004815599      4.321862e-05      1.426519e-04      0.0002153103
## medio          0.0003152952      4.450851e-05      9.610759e-05      0.0001335781
##      HouseStyleSFoyer HouseStyleSLvl OverallQual OverallCond YearBuilt
## economico      0.0002532111      0.0002550265      0.0007876906      0.0009062944      0.01682424
## medio          0.0002008359      0.0002250583      0.0007470030      0.0006247071      0.01280595
##      YearRemodAdd RoofStyleGable RoofStyleGambrel RoofStyleHip
## economico      0.013062159      0.0001202384      5.627251e-05      0.0001271590
## medio          0.008334655      0.0002312783      6.563805e-05      0.0001594939
##      RoofStyleMansard RoofStyleShed RoofMatlCompShg RoofMatlMembran
## economico      3.920651e-05      1.501972e-06      5.532289e-05      2.400228e-07
## medio          5.602619e-05      1.622848e-06      8.081182e-05      2.845022e-07
##      RoofMatlMetal RoofMatlRoll `RoofMatlTar&Grv` RoofMatlWdShake
## economico      1.259127e-07      7.915787e-06      1.377532e-05      4.081561e-05
## medio          1.511586e-07      2.430316e-06      1.142554e-05      5.998133e-05
##      RoofMatlWdShngl Exterior1stAsphShn Exterior1stBrkComm

```

```

## economico      4.787567e-06      4.557883e-07      3.993340e-06
## medio          2.371811e-05      1.727826e-07      1.871246e-06
##               Exterior1stBrkFace Exterior1stCBlock Exterior1stCemntBd
## economico      6.409669e-05      2.551975e-05      0.0003734905
## medio          8.981073e-05      2.820260e-05      0.0002954096
##               Exterior1stHdBoard Exterior1stImStucc Exterior1stMetalSd
## economico      0.0002849773      6.037610e-06      0.0003132943
## medio          0.0003455778      6.222043e-06      0.0002011090
##               Exterior1stPlywood Exterior1stStone Exterior1stStucco
## economico      7.472799e-05      4.665611e-05      6.002219e-05
## medio          6.455129e-05      5.022823e-05      7.041421e-05
##               Exterior1stVinylSd `Exterior1stWd Sdng` Exterior1stWdShing
## economico      0.0002587044      0.0002775036      7.058473e-05
## medio          0.0003130797      0.0001228940      7.230134e-05
##               Exterior2ndAsphShn `Exterior2ndBrk Cmn` Exterior2ndBrkFace
## economico      1.387971e-06      2.479224e-05      3.718113e-05
## medio          1.346984e-06      2.624154e-05      2.808158e-05
##               Exterior2ndCBlock Exterior2ndCmentBd Exterior2ndHdBoard
## economico      2.551975e-05      0.0003730714      0.0002769573
## medio          2.820260e-05      0.0002850796      0.0003147095
##               Exterior2ndImStucc Exterior2ndMetalSd Exterior2ndOther
## economico      1.637478e-05      0.0003236526      2.865251e-09
## medio          1.604781e-05      0.0002029960      6.542195e-06
##               Exterior2ndPlywood Exterior2ndStone Exterior2ndStucco
## economico      0.0001980831      4.858887e-05      5.343224e-05
## medio          0.0000877670      6.306255e-05      6.032283e-05
##               Exterior2ndVinylSd `Exterior2ndWd Sdng` `Exterior2ndWd Shng`
## economico      0.0002541827      2.331461e-04      0.0001438151
## medio          0.0003179308      8.282782e-05      0.0001137529
##               MasVnrTypeBrkFace MasVnrTypeNone MasVnrTypeStone MasVnrArea
## economico      0.0003416164      0.0003874761      8.237495e-05 0.09301635
## medio          0.0002361880      0.0003194825      1.433766e-04 0.11610110
##               ExterQualFa ExterQualGd ExterQualTA ExterCondFa ExterCondGd
## economico      9.284458e-05 0.0002583483 0.0003143904 8.480013e-05 1.124003e-04
## medio          3.317203e-05 0.0003969662 0.0002707649 1.119424e-05 9.261471e-05
##               ExterCondPo ExterCondTA FoundationCBlock FoundationPConc
## economico      4.066483e-05 1.319110e-04 0.0002392530 0.0002545884
## medio          4.559186e-07 7.243108e-05 0.0002452191 0.0002754405
##               FoundationSlab FoundationStone FoundationWood BsmtQualFa
## economico      5.840562e-05 2.706162e-05 3.149216e-06 8.448173e-05
## medio          3.728269e-05 4.025096e-06 9.895527e-07 1.467340e-04
##               BsmtQualGd BsmtQualTA BsmtCondGd BsmtCondPo BsmtCondTA
## economico      0.0004134461 0.0004058647 2.711733e-05 2.640644e-05 0.0000718554
## medio          0.0004588052 0.0003120336 1.318607e-04 5.689519e-08 0.0001297373
##               BsmtExposureGd BsmtExposureMn BsmtExposureNo BsmtFinType1BLQ
## economico      0.0002380998 0.0001190622 0.0004469458 9.097208e-05
## medio          0.0001378178 0.0001839419 0.0002793317 7.714111e-05
##               BsmtFinType1GLQ BsmtFinType1LwQ BsmtFinType1Rec BsmtFinType1Unf
## economico      0.0003384430 0.0001584538 0.0002036833 0.0001588483
## medio          0.0003383033 0.0001117593 0.0001416001 0.0001201403
##               BsmtFinSF1 BsmtFinType2BLQ BsmtFinType2GLQ BsmtFinType2LwQ
## economico      0.03417969 1.140216e-04 5.357128e-05 8.769492e-05

```

```

## medio      0.04368389      7.548475e-05      7.575708e-05      7.644901e-05
##      BsmtFinType2Rec BsmtFinType2Unf BsmtFinSF2 BsmtUnfSF TotalBsmtSF
## economico  0.0002638790      0.0002267775 0.05515344 0.02958684 0.05229199
## medio      0.0003629019      0.0002364323 0.07152798 0.03903486 0.07320409
##      HeatingGasA HeatingGasW HeatingGrav HeatingOthW HeatingWall
## economico 4.977862e-05 4.181255e-05 1.297872e-05 6.809691e-06 2.362325e-05
## medio      3.045633e-05 1.709898e-05 1.839082e-06 7.956691e-06 1.024473e-05
##      HeatingQCFA HeatingQCgd HeatingQCPo HeatingQCTA CentralAirY
## economico 9.998353e-05 0.0002683178 1.132878e-05 0.0003166748 0.0002945485
## medio      3.521856e-05 0.0001672985 8.816188e-06 0.0003830492 0.0000846418
##      ElectricalFuseF ElectricalFuseP ElectricalMix ElectricalSBrkr
## economico 1.055291e-04 9.675324e-06 4.302592e-08 0.0003616570
## medio      5.824351e-05 9.796053e-06 1.933887e-11 0.0001823056
##      X1stFlrSF X2ndFlrSF LowQualFinSF GrLivArea BsmtFullBath
## economico 0.05732055 0.04601704 0.09629693 0.04403392 0.0002780259
## medio      0.07715615 0.04983829 0.08793452 0.04828416 0.0002176794
##      BsmtHalfBath FullBath HalfBath BedroomAbvGr KitchenAbvGr
## economico 0.0001917648 0.0002489269 0.0002537577 0.0008718895 2.142726e-04
## medio      0.0001916004 0.0001308022 0.0002864086 0.0007487350 9.736856e-05
##      KitchenQualFa KitchenQualGd KitchenQualTA TotRmsAbvGrd FunctionalMaj2
## economico 1.011879e-04 0.0003661255 0.0004653296 0.0011797964 2.837907e-07
## medio      5.142075e-05 0.0003053397 0.0003526669 0.0008085687 4.273614e-08
##      FunctionalMin1 FunctionalMin2 FunctionalMod FunctionalSev
## economico 5.742257e-05 5.098698e-05 6.623612e-05 1.488455e-07
## medio      5.146807e-05 2.299805e-05 1.076459e-04 2.491323e-07
##      FunctionalTyp Fireplaces FireplaceQuFa FireplaceQuGd FireplaceQuPo
## economico 0.0001234834 0.0005149597 1.950680e-05 0.0002912886 3.682474e-05
## medio      0.0001404665 0.0002073631 1.175888e-05 0.0002146868 1.526448e-05
##      FireplaceQuTA GarageTypeAttchd GarageTypeBasment GarageTypeBuiltIn
## economico 0.0002793807 0.0003430456 0.0001780008 1.08285e-04
## medio      0.0001758973 0.0002762439 0.0002294329 9.96254e-05
##      GarageTypeCarPort GarageTypeDetchd GarageYrBlt GarageFinishRfn
## economico 9.24599e-05 0.000325671 0.01139246 0.0002645142
## medio      1.58306e-05 0.000317826 0.01137765 0.0002272131
##      GarageFinishUnf GarageCars GarageArea GarageQualFa GarageQualGd
## economico 0.0001957925 0.0002702174 0.07561051 1.004408e-04 5.317039e-05
## medio      0.0002902356 0.0003751510 0.11276245 6.228461e-05 5.049581e-05
##      GarageQualPo GarageQualTA GarageCondFa GarageCondGd GarageCondPo
## economico 7.286251e-07 1.435991e-04 1.071077e-04 5.507786e-05 6.378271e-05
## medio      9.161089e-08 5.131837e-05 4.905453e-05 1.344296e-05 5.893577e-06
##      GarageCondTA PavedDriveP PavedDriveY WoodDeckSF OpenPorchSF
## economico 1.797251e-04 4.588999e-05 1.447329e-04 0.1218680 0.1917451
## medio      4.789179e-05 3.132719e-05 6.906853e-05 0.1696755 0.2796749
##      EnclosedPorch X3SsnPorch ScreenPorch PoolArea PoolQCFA
## economico 0.1459642 0.002136383 0.2328127 0.03136260 5.349700e-05
## medio      0.1181985 0.002333017 0.2788786 0.03109531 5.470326e-05
##      PoolQCgd FenceGdWo FenceMnPrv FenceMnWw MiscFeatureOthr
## economico 5.431176e-05 5.774394e-05 1.116181e-04 2.051841e-05 3.792663e-05
## medio      5.604732e-05 2.522631e-05 3.853601e-05 8.890498e-06 4.841111e-06
##      MiscFeatureShed MiscFeatureTenC MiscVal MoSold YrSold
## economico 4.054199e-05 5.310364e-05 0.09422977 0.0017109937 0.01603303
## medio      5.562294e-05 5.439328e-05 0.13373654 0.0009669727 0.01532265

```

```

##          SaleTypeCon SaleTypeConLD SaleTypeConLI SaleTypeConLw SaleTypeCWD
## economico 8.516569e-07 5.422627e-05 4.483628e-06 1.280649e-05 4.034112e-05
## medio     8.020523e-07 4.351956e-06 7.011543e-06 1.269163e-05 1.370041e-05
##          SaleTypeNew SaleType0th SaleTypeWD SaleConditionAdjLand
## economico 8.129496e-05 3.272053e-06 0.0003539961 2.217397e-05
## medio     1.832100e-04 6.004988e-07 0.0002896359 1.051574e-05
##          SaleConditionAlloca SaleConditionFamily SaleConditionNormal
## economico 2.561715e-05 4.786098e-05 0.0002803497
## medio     1.884204e-05 1.498812e-05 0.0002716672
##          SaleConditionPartial SalePrice economico medio
## economico 8.667338e-05 0.0003557239 3.817706e-05 0.0000414717
## medio     1.831999e-04 0.0004087076 1.180959e-04 0.0002148406
##          caro
## economico 4.667592e-05
## medio     1.167636e-04
##
## Residual Deviance: 0.1498268
## AIC: 984.1498

```

Confusion Matrix and Statistics

```

##
##          Reference
## Prediction caro economico medio
## caro      8          0      0
## economico 2         1010    0
## medio     0          0    439
##

```

Overall Statistics

```

##
##          Accuracy : 0.9986
##          95% CI : (0.9951, 0.9998)
##          No Information Rate : 0.6923
##          P-Value [Acc > NIR] : < 2.2e-16
##
##          Kappa : 0.9968
##
##          McNemar's Test P-Value : NA
##

```

Statistics by Class:

```

##          Class: caro Class: economico Class: medio
## Sensitivity          0.800000          1.0000          1.0000
## Specificity          1.000000          0.9955          1.0000
## Pos Pred Value        1.000000          0.9980          1.0000
## Neg Pred Value        0.998622          1.0000          1.0000
## Prevalence            0.006854          0.6923          0.3009
## Detection Rate        0.005483          0.6923          0.3009
## Detection Prevalence  0.005483          0.6936          0.3009
## Balanced Accuracy      0.900000          0.9978          1.0000

```

Primero que nada, se observa que el modelo pudo predecir como se puede observar un porcentaje de 99.68%. Las predicciones para las clases economico y medio son casi perfectas, mientras que la clase caro muestra una leve disminución en la sensibilidad (80%), pero sigue siendo un buen resultado. La alta precisión y Kappa indican que el modelo está sobreajustado.

11. Compare la eficiencia del modelo anterior con los de clasificación de las entregas anteriores ¿Cuál se demoró más en procesar? ¿Cuál se equivocó más? ¿Cuál se equivocó menos? ¿por qué?

```
## [1] "Training time: 44.9276211261749"
```

De la entrega 2 del proyecto 2 se observa en terminos de tiempo de ejecución que este es el algoritmo más tardado a comparación de los modelos cruzados y no cruzados en las cuales toman un rango de tiempo de 2 a 10 segundos. En cuanto a los modelos que más se equivocaron se observa que el modelo knn tiene un porcentaje de presición de 74.66%. Aunque se ve que hay otros modelos que estan cerca del rango de 90% a 100% como Bayes se observa que este es el mejor modelo.